

Mapping Where Rumson Floods: A Sea-Level-Rise Simulation of the Navesink and Shrewsbury Watershed

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Abstract

Rumson, New Jersey is low-lying land between the Navesink and Shrewsbury rivers, and the town keeps no record of where it floods. A study area was traced in Google Earth and a United States Geological Survey (USGS) 3D Elevation Program (3DEP) digital elevation model (DEM) was read into R: 64.849 square kilometres (km²) at about 7 metres (m) per cell. A bathtub rule floods every cell at or below the new sea level; a second pass keeps only water with a path from the rivers. At +1 foot 214.6 acres are flooded, 1.6 percent of today's dry land, and at +5 feet 1,919.6 acres, 14.4 percent. Height, not distance from the shore, determines which land floods; the simple rule overstates the area by 14.4 percent at +1 foot and 6.8 percent at +5 feet. Shares are of dry land inside the mapped polygon, which also holds the Navesink Highlands, not of the Borough of Rumson. The connected acres are the values recommended for the town.

Introduction

This project focuses on how sea level rise could affect flooding in Rumson, NJ. Rumson is a low lying coastal community bordered by the Navesink and Shrewsbury rivers, making it vulnerable to rising water levels and more frequent coastal flooding. Understanding which areas are at the most risk can help residents and local officials make informed decisions about future resilience to ensure that the areas are prepared. This research aims to answer the question : How would different levels of sea level rise (1-5 feet) affect flooding patterns in Rumson, and which streets and areas would become vulnerable first? The data used is real geographic data for Rumson, NJ. The study area (Fig. 1) includes the Navesink River and the Shrewsbury River as well as the neighborhoods surrounding these rivers and other nearby coastal areas, such as the West Park peninsula (which has a low elevation and is an area of concern). The work was carried out during an internship with the Rumson Environmental Commission. The town keeps no flood records, yet people already see flooding on calm days at high tide.

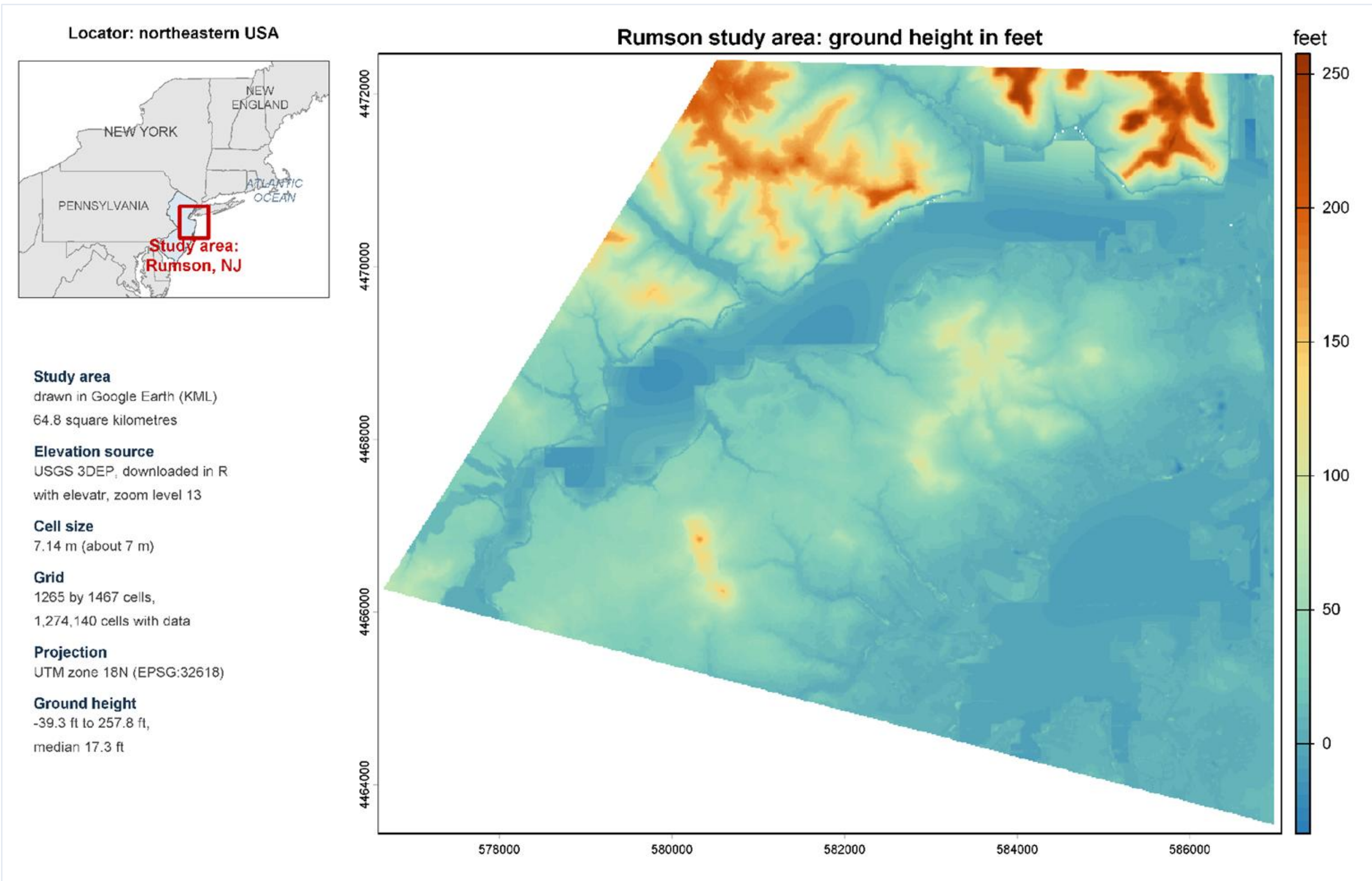


Fig. 1. Locator map of the northeastern United States beside a colour map of ground height in feet across the study area (USGS 3DEP). The red box is a marker, not the true footprint; the Navesink Highlands reach 257.8 feet (ft). This mapping provides context for understanding why lower-lying coastal regions experience more rapid flooding compared to higher inland terrain.

Methodology

- Draw. Polygon traced in Google Earth, exported as Keyhole Markup Language (KML): 64.849 km² (Fig. 1).
- Download. The R package elevatr (elevation data in R) pulls USGS 3DEP heights in metres, `get_elev_raster(z = 13)`.
- Project and convert. Universal Transverse Mercator (UTM) zone 18N, European Petroleum Survey Group (EPSG) code 32618, clipped to the polygon, then metres + 0.3048 into feet.
- Check and clean. The DEM is 1265 by 1467 cells at 7.14 m, 1,274,140 with data, ~2,071 ft to 257.8 ft. The 161 cells below -40 ft (0.0126 percent) set to no data: map colour only.
- Bathtub rule. Flooded at level L = ground above 0 ft and at or below L, L = 1 to 5 ft. Acres = cells × 50.98 square metres (m²) ÷ 4,046.86 m².
- Test the rule. `patches()`, in the R package terra, keeps only water joined to today's water.

Results

- Nonlinear trend, shown in fig. 4 through the acceleration of land lost as sea level rises in increasing percentages
- Every extra foot takes more land: 1.6 percent of today's dry land at +1 ft, 14.4 percent at +5 ft.
- The share of dry land flooded rises in steps of 1.6, 2.6, 3.0, 3.5 and 3.7 percentage points, so the increments grow rather than stay equal.
- Counting only water connected to the rivers gives a smaller flooded area than the bathtub rule at every level (Table 1).
- Water follows both rivers and the low eastern side; the Highlands, 257.8 ft, stay dry (Fig. 2).
- At +1 and +2 feet sea level rise, the red areas are thin and only around immediate shoreline areas and low lying edges, and at +3, +4, and +5 feet sea level rise, the red areas are spread significantly outward to deeper inland across the perimeter of the study area (Fig. 2)
- Fig. 3 shows that requiring a direct water pathway from the rivers reduces the calculated +3 foot flooded area by 11.3 percent because unreachable inland pockets are excluded

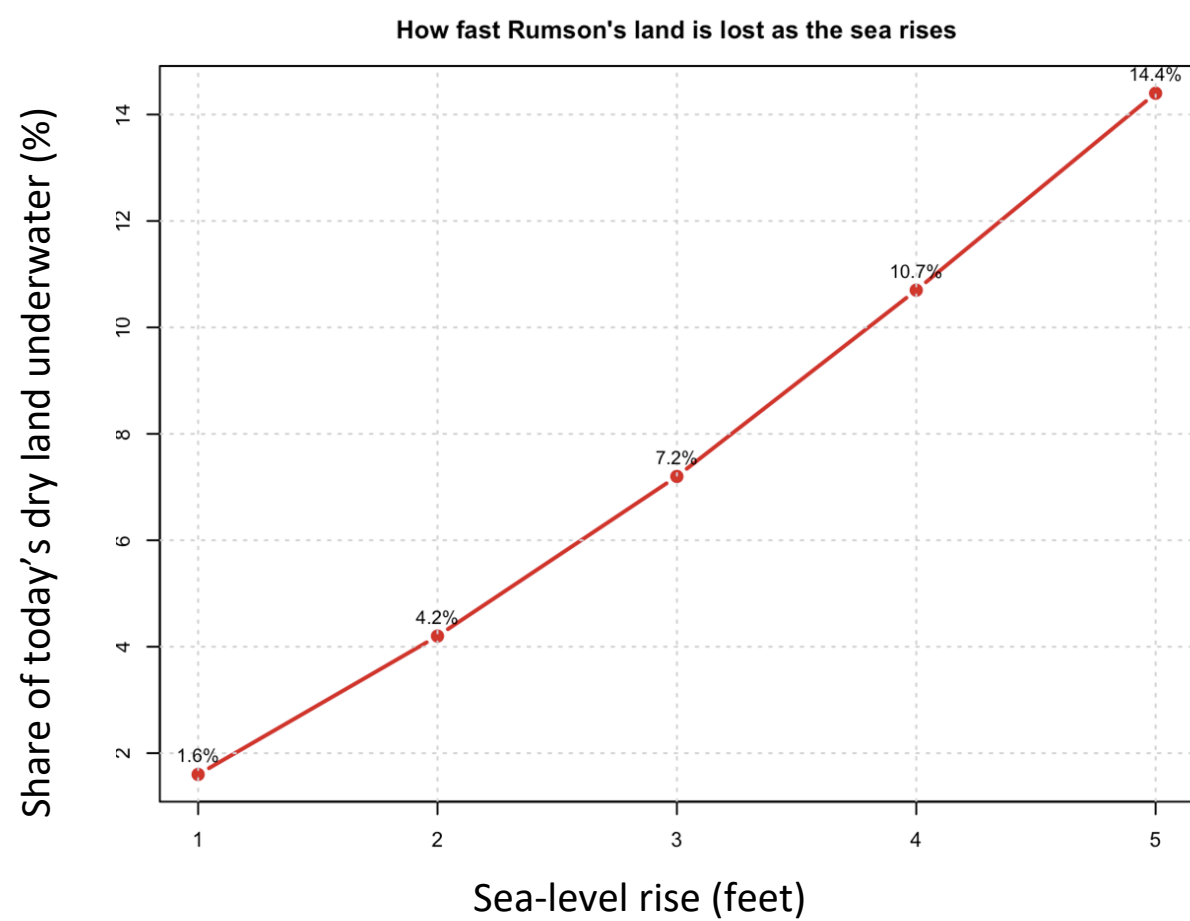


Fig. 4. Share of the dry land inside the study polygon that goes under water at each level of sea-level rise, from +1 to +5 feet. The steps between levels grow rather than stay equal.

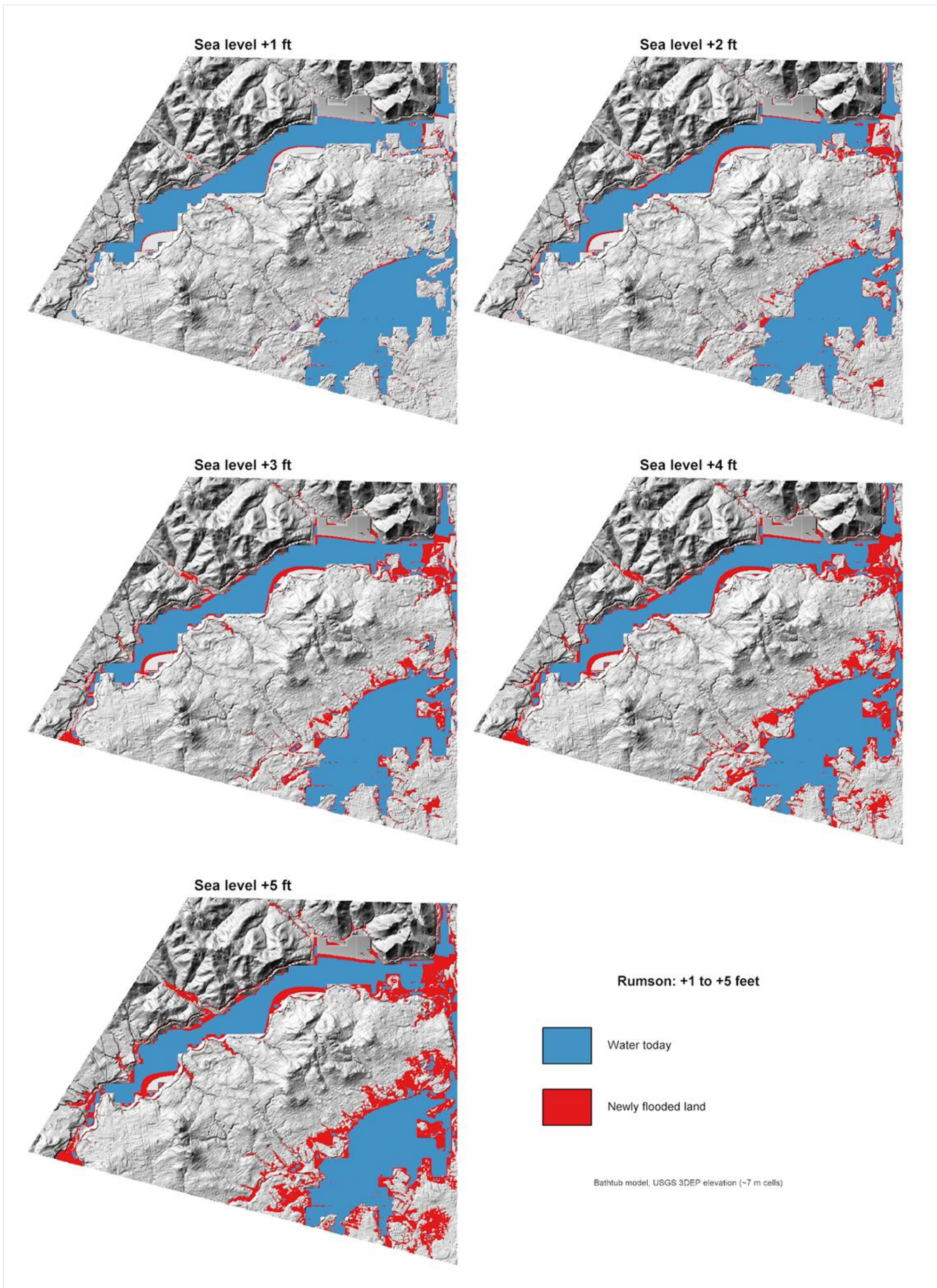


Fig. 2. Five maps of simulated flooding at +1 to +5 feet of sea-level rise: blue is water today, red is land dry today and under water at that level. Red grows along both rivers and the low east; the Highlands stay dry. Flooding concentrates along the Navesink and Shrewsbury rivers and the low eastern neighborhoods, while the Navesink Highlands stay dry.

Table 1. Flooded land at each rise: bathtub acres, connected acres, and the gap. Percentages are shares of dry land in the 64.849 km² study polygon, which also holds the Navesink Highlands, not of the Borough of Rumson.

Sea-level rise	Bathtub acres	% of dry land	Connected acres	Overstated by
+1 ft	214.6	1.6 %	183.8	14.4 %
+2 ft	560.1	4.2 %	487.4	13.0 %
+3 ft	963.4	7.2 %	854.8	11.3 %
+4 ft	1,421.0	10.7 %	1,284.2	9.6 %
+5 ft	1,919.6	14.4 %	1,789.0	6.8 %

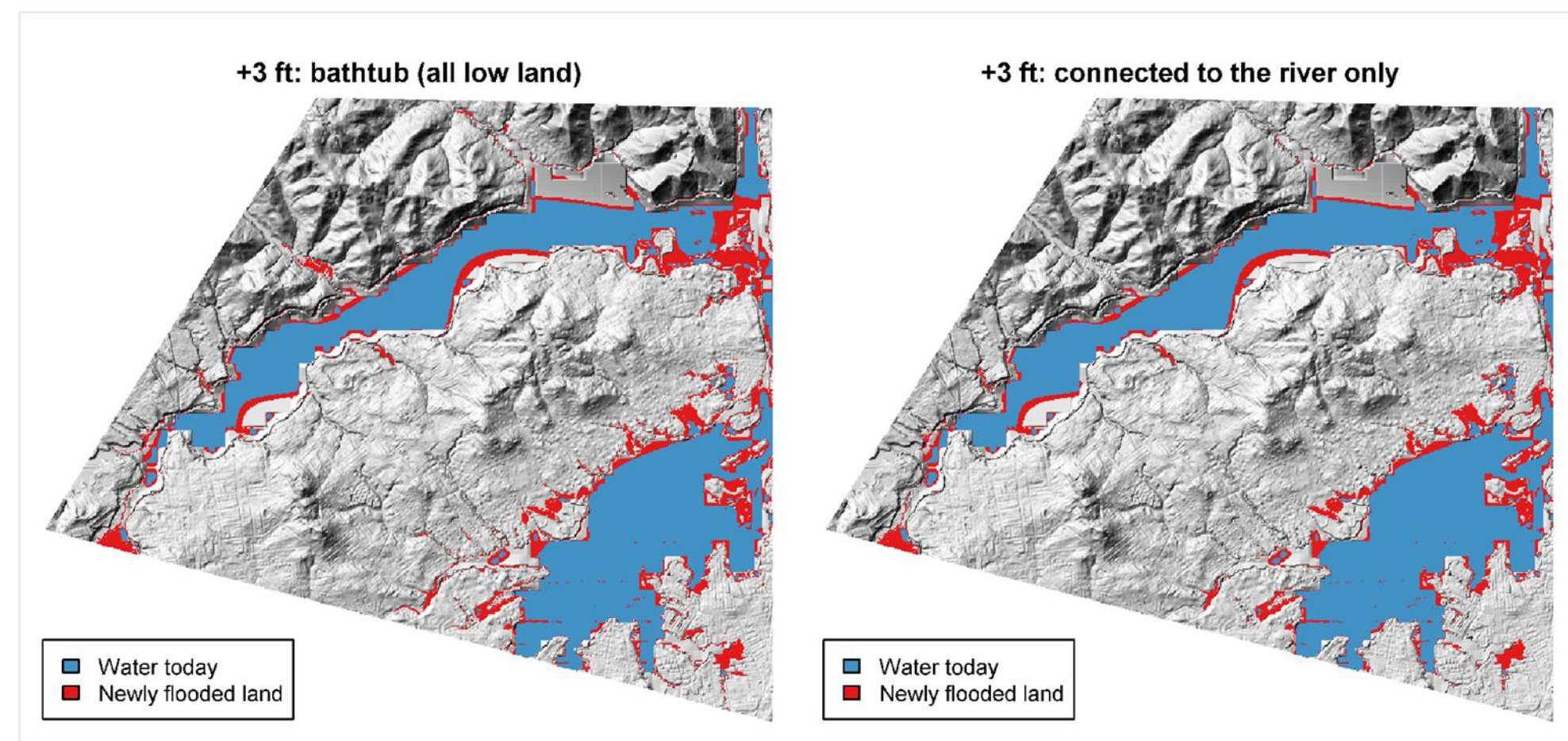


Fig. 3. Two maps of the same +3 foot flood: the bathtub rule on the left, only water with a real path from the rivers on the right. Dropping unreachable pools shrinks the flooded area by 11.3 percent (Table 1).

Discussion and Conclusions

- The grid has no road layer, so the individual streets in the research question cannot be identified.
- Height, not distance from shore, determines which land floods. The Highlands stay dry at +5 ft.
- The connected acres in Table 1 are the values recommended for the town, and the simple map is weakest for the small, early rises.
- Topography : The elevation models show that Rumson's terrain is made up of low lying coastal areas along the Navesink and Shrewsbury Rivers
- Initial vulnerability : Shoreline zones with low elevation are directly exposed, making them primary entry points for flooding as sea level rises
- Simulated flooding in the Rumson study area for 1 to 5 feet of sea-level rise :
 - Relationship is not linear ; as elevation gets higher, each added foot of sea level rise floods a larger volume of newly submerged land than the foot before it
 - +1 Foot : immediate shoreline areas are impacted
 - +2 Feet : flooding is expanded onto waterfront properties
 - +3 Feet : flooding expansion onto low lying residential neighborhoods
 - +4 Feet : significant broadening of the flood zone is a result of flooding
 - +5 Feet : severe flooding across the perimeter and low elevation land of Rumson
- Main takeaway : Sea level rise in Rumson is not gradual or steady - it is accelerating and threatening. Understanding this acceleration and risk can lead to informed decisions about the future of these coastal communities. For example, informed decisions about infrastructure planning, property awareness, implementing projects that absorb rising water, and guiding smarter choices to protect these communities for generations to come.

LIMITATIONS

A height rule, not a flow model: no tides, waves, surge, rain, groundwater, pumps, sea walls or berms, and no timeline, so +1 to +5 ft is not tied to any year. Bare ground at about 7 m per cell cannot see a street or a bulkhead. The North American Vertical Datum of 1988 (NAVD88), the model's vertical zero, is not the local tidal datum.

FUTURE WORK, NOT PART OF THESE RESULTS

The tidal side is not done. USGS gauge 01407600, Shrewsbury River at Sea Bright, reports tidal elevation (parameter 72279, NAVD88, in feet) every 6 minutes. One raw record was plotted; the tide has not been analysed and no forecast was built. Federal Emergency Management Agency (FEMA) flood zones and the National Oceanic and Atmospheric Administration (NOAA) viewer were background reading only, never brought into R. Also, findings will be presented to urge community members to take action to implement nature based solutions and ensure long term coastal resilience and interactive web map will be developed so community members can zoom in to specific neighborhoods to explore the impact on sea level rise to them individually in the area of their own home

SIGNIFICANCE

Rumson has no written record of where it floods. This is the first map of the town built from measured ground heights, and it can be given to the Rumson Environmental Commission with a measure of how much the bathtub rule overstates the flooded area.

SELECTED REFERENCES

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KEYWORDS AND CONTACT

sea-level rise, flood mapping, bathtub model, hydrological connectivity, digital elevation model, USGS 3DEP, R, Rumson New Jersey
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